DermAssist AI — Clinical Feedback Packet

Audience: Doctors, dermatologists, clinics, hospitals (non-technical overview)
Purpose: Explain what this research prototype is, what it does **not** do, and how we are seeking **professional perspective**—not patient care decisions based on this tool.

Important: DermAssist AI is **not** a diagnostic device. It has **not** been reviewed by FDA or similar regulators for medical use. It is **not** approved for clinical deployment.

What is DermAssist AI?

DermAssist AI is a **research demonstration** that combines:

- A simple web upload screen.
- A computer model (**EfficientNet-B0**) trained on the **public HAM10000** skin lesion image collection.

The system produces a **research-style screening label** ("Benign" vs "Potentially Malignant") and may show a **colored heatmap** indicating where the model focused visually. It is meant for **discussion, education, and workflow feedback—not** for diagnosing patients.

What does the model actually do?

The model looks at **patterns in the image pixels**—texture, color, and structure—that it learned from thousands of **labeled research images**. It compares a new image to patterns seen during training and outputs:

- A **category** (research grouping described below).
- A **confidence score** that reflects how strongly the model's math favors one category over the other—not clinical certainty.

This is **pattern recognition on photographs**, not an examination, not a biopsy, and not a substitute for clinical judgment.

What "Benign" vs "Potentially Malignant" means here

These labels come from a **research grouping** of public dataset categories—not from a pathologist's diagnosis of your patient.

- **"Benign" (research label)** corresponds to a lower-risk group of diagnoses used in our training setup.

- **“Potentially Malignant” (research label)** corresponds to a higher-risk group used for screening-style experiments.

These names do not mean the lesion is truly benign or malignant in real life. They only describe how we grouped labels for research.

What the Grad-CAM heatmap shows

When a heatmap is displayed, it highlights areas of the image that **most influenced the model’s score** for its prediction. Think of it as “where the computer looked hardest”—**not** “where the cancer is,” and **not** proof that an area is abnormal.

Heatmaps can be influenced by lighting, blur, hair, rulers, or background skin. They should **never** replace dermoscopy skill or biopsy decisions.

What the confidence score means

The confidence score is a **technical measure** from the model’s output. A higher score means the model is **more decisive** between its labels—not that the finding is medically more certain or “more dangerous.” Low confidence does **not** guarantee safety, and high confidence does **not** prove disease.

What we need you to understand about mistakes

- The system **can be wrong**.
- **False negatives are possible:** the tool may suggest “Benign” when, in reality, a concerning lesion would warrant evaluation.
- **False positives are possible:** the tool may raise concern when clinical judgment would not.

Clinicians should not rely on this system for patient care. It does not replace history, full skin exam, dermoscopy training, or indicated biopsy and pathology.

How this relates to public datasets

The model was trained on **HAM10000**, a **public** set of dermoscopy images from past research. Real clinics see different cameras, skin tones, lesion types, and workflows. Performance on public data **does not** predict performance in your practice.

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1. Why we built this

We built DermAssist AI to:

- Show how modern AI **pipelines** can be connected to transparent explanations (heatmaps) and clear disclaimers.
- Invite **honest clinical feedback** on whether such tools could ever be useful in **controlled research**—and where they fail or cause harm if misunderstood.

We are **not** claiming this tool is ready for patients.

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2. Current capabilities

- Upload an image and receive a **research screening-style** label and confidence.
- Optionally view a **heatmap** of model attention.
- Read built-in reminders that output is **not** diagnostic.

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3. Current limitations

- **No clinical validation** in real practices or prospective trials.
- **Single public dataset** training—limited diversity and realism.
- **No external validation** on independent hospital data in this prototype's reporting.
- **No regulatory approval** for medical use.
- Heatmaps show **model behavior**, not confirmed pathology.

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4. What feedback we are requesting

We welcome professional input on:

- Whether the **language and disclaimers** are clear enough to reduce misunderstanding.
- Whether the **heatmap presentation** risks being interpreted as diagnostic "localization."
- What **evidence** would be needed before **any** future study involving clinicians or patients.
- Practical barriers (workflow, liability, imaging quality) even if the technology improved.

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5. What we are NOT requesting

- We are **not** asking you to use this tool to diagnose or treat patients.
- We are **not** asking you to endorse safety or efficacy for clinical use.
- We are **not** implying regulatory clearance or hospital readiness.

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6. Privacy statement

This research prototype is intended for **local or controlled demos**. If images are sent to a server, they should be handled under policies appropriate to your institution. **Do not upload real patient images** unless you have proper authorization, consent, and compliance review. For demonstrations, use **de-identified** or **public** example images only.

When in doubt, **do not** upload clinical data to external systems.

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7. Future research direction

Possible next steps—subject to ethics approval and regulation—include:

- Studies with **more diverse datasets** and **independent validation**.
- **Prospective** workflows with supervised use only.
- Collaboration with dermatologists on **safe presentation** of AI outputs.
- Clear separation between **research demos** and any future regulated product development.

Summary

DermAssist AI analyzes **visual patterns** learned from a **public** skin lesion dataset and produces **research screening-style** outputs and optional attention maps only. It is **not** a diagnosis, **not** FDA-approved, and **must not** guide patient care. We seek **professional critique** to keep expectations realistic and safety central.

Placeholder — product screenshot: *[Insert neutral demo screenshot here — no patient identifiable information.]*

Placeholder — example heatmap: *[Insert example heatmap with caption: "Model attention only—not pathology."]*

For technical detail, see `docs/final_project_report.md` and `docs/model_evaluation.md`. For full disclaimer text, see `docs/disclaimer.md`.